CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

December 2013

CL307 Structural Analysis-III

Date: 04.12.2013, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 A concrete beam with a cross-sectional area of 32×10^3 mm^2 and radius of gyration of 72 mm is prestressed by a parabolic cable carrying an effective stress of 1000 N/mm^2. The span of the beam is 8 m. The cable, composed of 6 wires of 7 mm diameter, has an eccentricity of 50 mm at the centre and zero at the supports. Neglecting all losses, find the central deflection of the beam as follows: **[07]**

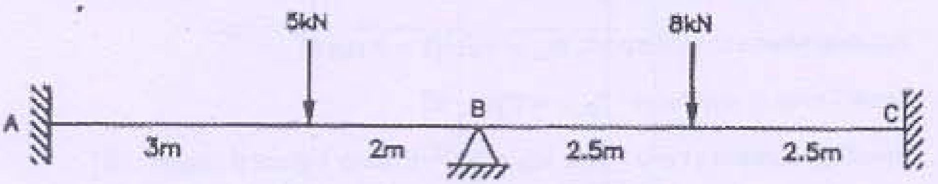
(a) Self-weight + prestressing force

(b) Self-weight + prestressing force + love load of 2 kN/m.

Q - 2 (a) A rectangular concrete beam, 100 mm wide by 250 mm deep, spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm. The beam supports a live load of 1.2 kN/m. Calculate the resultant stress distribution for the central cross section of the beam. The density of concrete is 24 kN/m^3. **[04]**

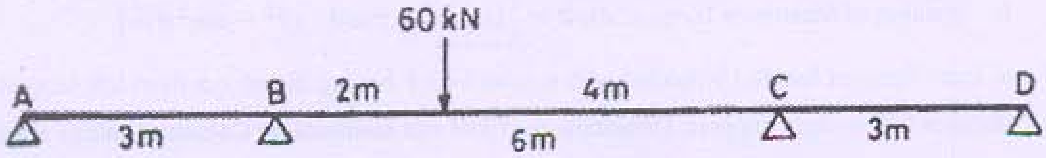
(b) Using slope deflection method, determine the support moments and reactions for the beams shown in Figures below. Draw also S.F. and B.M. diagrams. **[Attempt any One]** **[10]**

(i)



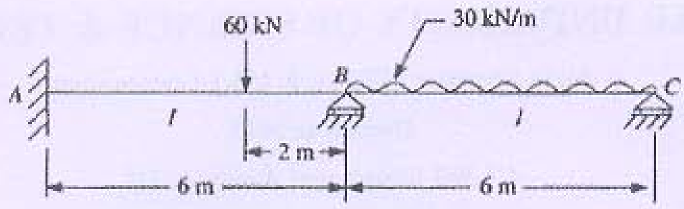
EI=CONSTANT

(ii)

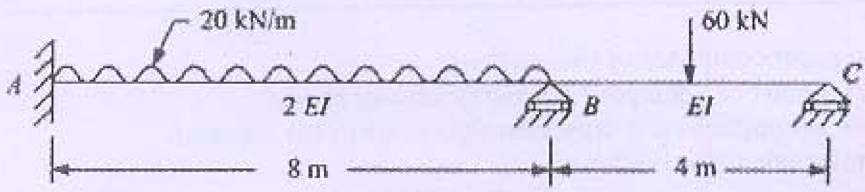


Q - 3 Using moment distribution method, determine the support moments and reactions for the beams shown in Figures below. Draw also S.F. and B.M. diagrams. **[Attempt any Two]** **[14]**

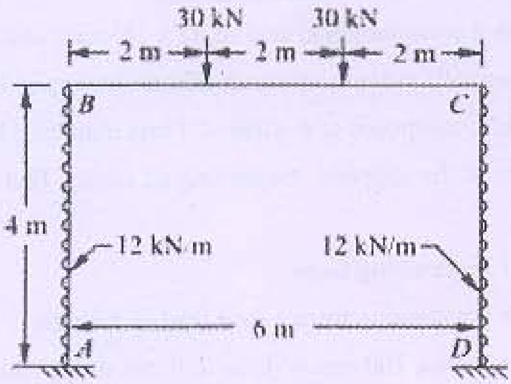
(i)



(ii)



(iii)



SECTION – II

Q – 4 A beam circular in plan is located uniformly with U.D.L. of 100 kN/m inclusive of self-weight. The radius of beam is 4.5m. The beam is symmetrically supported on five columns. Draw S.F., B.M. and T.M. (Use relevant equation from below) **[07]**

1. Shear Force at Support: $F_0 = wR\theta$
2. Bending Moment at Support: $M_0 = wR^2[1 - \theta \cot\theta]$
3. Shear Force at any Point : $F_{\emptyset} = wR[\theta - \emptyset]$
4. Bending Moment at any Point: $M_{\emptyset} = wR^2[\theta \sin\emptyset + \theta \cot\theta \,.\cos\emptyset - 1]$
5. Twisting Moment at any Point: $T_{\emptyset} = wR^2[\theta \cos\emptyset - \theta \cot\theta \,.\sin\emptyset - (\theta - \emptyset)]$
6. Position of Maximum Torque: $sin\ \emptyset = \frac{1}{\theta}[sin^2\theta \pm \cos\theta \,.\ (\theta^2 - sin^2\,\theta)^{\frac{1}{2}}]$

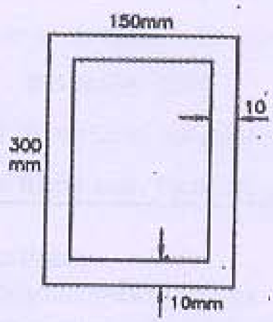
Q - 5 (a) A fixed beam of length l is loaded with a point load P having distance a from left support and distance b from right support. Determine the fixed end moments by Column Analogy method. **[04]**

(b) Attempt any Two. **[10]**

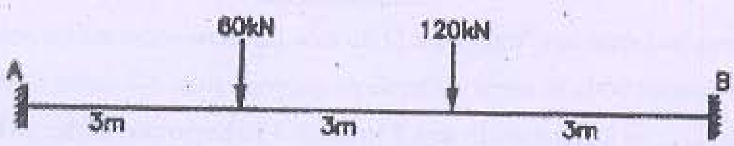
(i) Find out shape factor for a T Beam having Flange dimensions 150mm x 10mm connected with a web of dimensions 200mm x 10mm.

(ii) Determine the shape factor and plastic moment of resistance of the given section. Take F_y=250

N/mm^2.

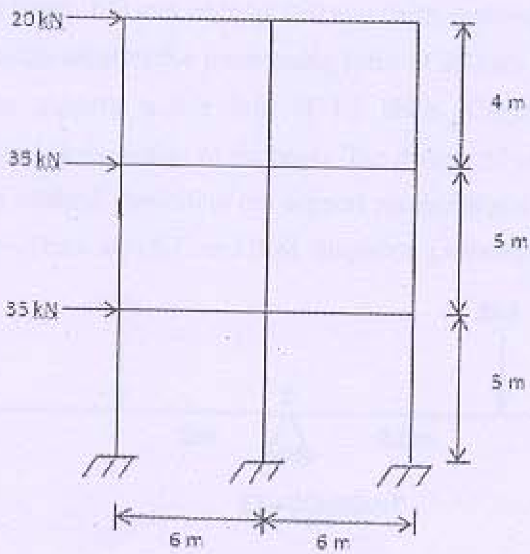


(iii) Compute fully plastic moment for a beam shown in Figure.



Q - 6 (a) A fixed beam of length l carries udl of intensity w kN/m acting for left half length. Determine the fixed end moments by Column Analogy method. **[05]**

(b) Analyse the portal frame shown in Figure by portal/cantilever method. **[09]**



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